MILITARY SPECIFICATION

REPAIR SET, AIRCRAFT A/E24M-41

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable, self-contained aircraft hot bonded repair set with capability to generate, control, and monitor vacuum pressures and cure temperature. This repair set is to be used in the repair and dry-out of aircraft bonded and composite structures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-8512	Support Equipment Aeronautical Special, General Specification for the Design Of
MIL-P-13949	Plastic Sheet, Laminated, Metal Clad (for Printed Wiring Boards), General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4920

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MIL-P-13949	Plastic Sheet, Laminated, Metal Clad (for Printed WiringBoards), General Specification For
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification for
MIL-C-28809	Circuit Card Assemblies, Rigid, Flexible, and Rigid-Flex
MIL-G-43976	Gloves and Glove Set, Protective
MIL-I-45208	Inspection System Requirements
MIL-P-50884	Printed Wiring, Flexible and Rigid-Flex
MIL-P-55110	Printed Wiring Boards, General Specification For
MIL-P-83497	Procedures for Preparation of Programmed Tapes and Cards
MIL-M-87929	Manuals, Technical: Maintenance Instructions in Work Package Format for Engines (Intermediate and Depot)

STANDARDS

FEDERAL

FED-STD-595	Colors Used in Government Procurement

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-721	Definitions of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support and Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-45662	Calibration Systems Requirements
MS 3408	Connectors, Plug, Electric, 90° Assembly, Crimp Contact, AN Type
MS 3451	Connectors, Receptacle, Electrical, Cable Connecting, Rear Release, Crimp Contact, AN Type
MS 3452	Connectors, Receptacle, Electrical, Box Mounting, Rear Release
MS 3456	Connectors, Plug, Electrical, Rear Release, Crimp Contact, AN Type

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D; 700 Robbins Ave., Philadelphia PA 19111-5094.)

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2.2 Non-Government publication. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ANSI/ASME B40.1 Gauges-Pressure Indicating Dial Type-Elastic Element (DoD adopted)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th St., New York NY 10017-2392.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia PA 19103-1187.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 7241-1 Hydraulic Fluid Power - Quick-Action Couplings - Part 1: Dimensions and Requirements

(Application for copies should be addressed to the International Organization for Standardization, 3 rue de Varembe, 1211 Geneve 20, Switzerland.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), two samples shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Item requirements. The bonded repair set provided shall be capable of automatically generating, controlling and monitoring vacuum pressure, cure time and temperatures in world wide use. Each set shall include heater blankets, temperature controllers, monitoring devices, a vacuum source, temperature indicators/recorders, and all necessary cables/connectors to perform part dry-out and composite/metal bonded repairs. The repair set shall contain two controllers that are capable

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of operating totally independent cures of different temperatures, cure times, start/stop times and post or step cures simultaneously. The repair set shall require only electrical power source to generate heat and vacuum for application to the area to be repaired (shop air or vacuum may be used when utilizing the auxiliary venturi or vacuum systems). The set shall include all components, parts and features necessary to meet the performance requirements specified herein included in two containers or less.

3.2.1. Interface requirements

3.2.1.1 Power. The repair set shall have the capability to operate using commonly available power supplies of 110-120 AC/60 Hz single phase and 220-240 AC/50 Hz single phase input power with a selector switch that is protected against inadvertent power application. Automatic voltage selection shall be considered as an alternative to a selector switch. The repair system shall be rated 30 amps at 120 AC and 20 amps at 240 AC. The repair set will accept power into the console using a box mounting receptacle which conforms to MS 3452, class W, size 20-19, style P, wall mount with an E backshell. Blanket power outlets shall be box mounting receptacles which conform to MS 3452, class W, size 22-28, style S, wall mount with an E backshell. Blanket power connectors shall utilize pins A and F for 110 volt operation and pins C and D for 220 volt operation.

3.2.1.1.1 Steady-state conditions. The repair set shall operate and maintain the specified performance and accuracy within a steady-state voltage tolerance band of $\pm$10 percent and within a steady-state frequency tolerance band of $\pm$5 percent with no detrimental effect on equipment performance.

3.2.1.1.2 Transient-state conditions. The repair set shall not experience alterations of characteristics due to the transient-state conditions as defined in MIL-T-28800 after a maximum transient time of 500 microseconds which recovers to within the steady-state tolerance band within 5 seconds. Operation for transient-state conditions which persist longer than 5 seconds shall be as required for interruption of source power.

3.2.1.1.3 Interruption of power. The repair set shall be capable of resuming normal operation following voltage or frequency interruptions which are greater than the transient values. When power interruptions of 2 minutes or less occur and after power is restored, the repair set shall automatically activate a recovery mode, detailed in paragraph 3.4.1.r. During and after power interruptions of greater than 2 minutes, the repair set shall be capable of normal operation after the power is restored.

3.2.1.2 Operating conditions. The repair set components shall be sized to permit set-up and operation by personnel wearing chemical/biological warfare gloves conforming to MIL-G-43976.

3.3 Design and construction

3.3.1 Materials, parts and processes. The materials and components used shall be new and only those materials and components determined by actual test performance to be of the highest quality. Materials, parts and processes selected shall be capable of passing the performance and test requirements of this document.

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3.3.1.1 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design.

3.3.1.2 Standard parts. Except as otherwise specified herein, MS, AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts and cotter pins may be used provided they have suitable properties and are replaceable by the MS, AN or MIL standard parts without alteration and provided the corresponding MS, AN, or MIL standard part numbers are referenced on the drawings and in the parts list.

3.3.1.3 Magnesium. Magnesium shall not be utilized in the construction of the repair set.

3.3.1.4 Dissimilar metals. Dissimilar metals in contact, as defined by MIL-STD-889, shall not be used.

3.3.1.5 Fungus resistance. Material selected shall be fungus resistant in accordance with MIL-STD-454.

3.3.1.6 Corrosion resistance. Materials selected shall be corrosion resistant or suitably protected in accordance with MIL-STD-808 for type I equipment.

3.3.1.7 Measuring and indicating devices. All measuring and indicating devices such as temperature controllers, temperature indicators, and all other similar devices shall be graduated in the Fahrenheit temperature scale.

3.4 Performance

3.4.1 Temperature controller. The repair set shall have 2 independent temperature controllers. Each controller shall have the following features:

a. The controller shall have an ON-OFF switch with a visible indication of power ON.

b. The controller shall be capable of automatically controlling temperatures up to 999 °F.

c. The controller shall be removable, repairable, and replaceable by a trained technician.

d. The controller shall be provided with a control thermocouple to monitor repair temperature and provide input to the controller. Auxiliary thermocouples shall be provided to each controller to monitor repair temperature in other locations. The controller shall insure continuity of all thermocouples.

e. The control thermocouple temperature shall be indicated and recorded. The thermocouple shall not be integral with the heater blanket.

f. The controller shall automatically trigger an audible and visible alarm when the control thermocouple temperature deviates from the set point temperature by ±15 °F. An audible and visual alarm shall also be indicated for any auxiliary thermocouple temperature (when used) that deviates

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from the set point temperature by ±25 °F. The controller shall automatically shut off power to the heater blanket if the control thermocouple temperature exceeds the set point temperature by +20 °F but not abort the program cycle. When the control thermocouple falls to +10 °F of the set point temperature, the programmed cycle shall resume.

g. The controller shall be programmed to meet the requirements specified in paragraph 3.4.1.1.

h. Temperature controller accuracy shall be plus or minus (1/4% of actual reading + 1 least significant digit).

i. The controller shall be capable of sensing system level vacuum and be provided with an adjustable, audible and visible high and low limit vacuum alarm.

j. The controller shall trigger an audible and visible indication at the end of curing cycle (when the final temperature cools to 120 °F) and remain on for a duration of 5 seconds.

k. The controller shall visually display current controller mode (e.g. program, run).

l. The controller, or repair set, shall visually indicate the progress of the programmed cure cycle.

m. The controller shall visually display the repair area temperature as monitored by the control thermocouple, and the programmed set point temperature at the same time.

n. All cure cycle parameters shall be displayed continuously and remain on until the controller is programmed for another cure cycle or shut down. When in a temperature dwell, the controller shall visually display time to complete the temperature dwell. Display and program variables shall be written in English (no symbols or codes).

o. The controller shall have a keypad with spring-loaded keys no smaller than .5 inch x .5 inch for minimization of operator error during programming. Key displacement shall be a minimum of .05 inch to a maximum of .25 inch. Key resistance shall be a minimum of 1.0 ounces to a maximum of 10.0 ounces.

p. All displays and indicators shall be visible outdoors in sunlight without hoods or special screens and indoors under less than 20 to more than 100 foot candles of illumination at the face of the instrument when viewed at an angle of 45° or less from the plane of the instrument face.

q. The audible alarm shall have a volume of 80 db aweighted sound pressure level at a three-foot distance. Associated with all alarms shall be a visual alarm indication of at least .50 inch length by .25 inch width in size. The audible alarm shall have a silence button, however, the visual alarm indicator shall remain on until the deviation has been eliminated. The audible alarm shall be able to be re-triggered for any additional deviations in this mode. The visual and audible alarms shall reset automatically when the deviation (s) is (are) corrected.

r. The controller shall activate a recovery mode when power has been interrupted for 2 minutes or less and then restored. The recovery mode operates as follows:

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(1) A new set point temperature is established from the control thermocouple indication.

(2) If the controller is to resume program operation of a ramp, the controller shall continue the ramp at the rate as programmed.

(3) If the controller is to resume program operation of a dwell, the controller shall ramp the repair temperature at the programmed rate used to achieve that dwell temperature. Dwell time is not advanced during this recovery ramp. When the dwell temperature is achieved, the dwell time is resumed.

3.4.1.1 Controller programmability. The temperature controller shall have the following features as a minimum:

a. The controller shall have the capability to perform single, double and triple step cures (A "step" consists of one ramp and one dwell cycle). The controller shall retain the last programmed cure cycle as default values.

b. Heat up and cool down rates shall be programmable from 1 °F/min to 15 °F/min.

c. The program shall only require the operator to input heat-up and cool-down rates, dwell temperatures and dwell times. All other program parameters shall be preset and not easily accessible to the operator.

d. When programming the cure cycle, the controller shall visually prompt the operator with directions to input the cure cycle. Program variables shall be written in English (no symbols or codes). An acceptable example is as follows:

(1) "Input number of steps in the cure. (1,2 or 3)"

(2) "Input ramp rate (°F/min) for step __."

(3) "Input the dwell temp (°F) for step __."

(4) "Input the dwell time (min) for step __."

(Previous three directions shall repeat for remaining steps in cure cycle.)

(5) "Input the cool-down rate in °F/min."

(Next, the controller shall printout and display the entire programmed cure cycle.)

(6) "Is cure cycle correct (Y/N)?"

(If answer is 'N', controller shall default to direction (1). If the answer is 'Y', the controller shall proceed.)

(7) "Press RUN key to begin cure cycle."

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e. The controller shall sense the ambient temperature of the repair area using the thermocouple on the repair area. This temperature shall be used by the controller as the initial set point.

f. The program shall be temperature sensitive so that the dwell times begin when the dwell temperature is achieved as monitored by the thermocouple as specified in 3.4.1.d.

g. The controller shall have non-volatile memory for a period of one year minimum.

3.4.1.2 Thermocouples. The repair set shall be provided with thermocouples which meet the following criteria:

a. Thermocouples shall be J-type and be accurate to within ±4 °F when assembled to the 50 foot length.

b. Standard size plugs (Marlin P/N 1060-J-P or equivalent) shall be utilized (no miniatures).

c. All connecting wires and leads shall be designed to eliminate air leak paths (i.e., fused and non-porous insulation and covers) during vacuum bagging operations and shall be installed using strain relief techniques.

d. Thermocouples shall not be integral to the heater blanket.

e. Thermocouples shall have welded tips.

3.4.1.3. Auxiliary temperature indicator. The repair set shall have a digital temperature indicator with the following features:

a. The indicator shall provide visual numeric display of at least 8 thermocouples separate to the controller input device.

b. Selector switches shall be provided to select input for read-out. Thermocouples will be assigned to a controller (4 minimum per controller) with the capability of "slaving" all thermocouples if required.

c. The auxiliary temperature indicator shall be capable of removal and installation by a trained technician.

d. The temperature indicator range shall have a 3 digit indication with a range of 000 to 999 °F.

e. The auxiliary temperature indicator accuracy shall be ±1/2% of the indicated reading or 1 °F, whichever is greater.

3.4.1.4 Temperature/vacuum/deviation alarm recorders. The repair set shall be provided with a temperature, vacuum, and deviation alarm recorder for each controller. Each recorder shall have the following features:

a. The recorder shall record temperature, vacuum, and deviation alarm messages on one output print.

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b. The recorder shall record temperature from the control thermocouple as monitored at the part surface and vacuum as monitored from the repair set.

c. The recorder shall be an alpha-numeric recorder using paper and/or ribbons commercially available.

d. The recorder shall be easily set and loaded without disassembly of the repair set. The recorder print interval shall be programmable with print intervals ranging from "OFF" to "10" minutes. This function may be integral or independent of the controller functions.

e. The recorder shall be capable of printing the following information as a minimum:

(1) Self test status.

(2) Cure cycle program (dwell temperatures and times, ramp and cooling rates and minimum vacuum alarm setting).

(3) Print interval information (ramp or dwell number, set point TEMP, actual TEMP and vacuum level).

(4) Deviation alarm information (time of incidence, deviation nomenclature and values, time of deviation correction).

(5) Cure cycle complete indication.

(6) Reprint of cure cycle program, if modified.

(7) All auxiliary thermocouple TEMP readings.

(8) Recovery mode including commencement time, action (new set point, ramp rate), and recovery mode completion.

f. The recorder shall require no special calibration.

g. The recorder shall be solid-state and removable without console disassembly.

3.4.1.5. Vacuum. The repair sets shall be capable of generating a vacuum of 27 inHg at sea level on a standard day and evacuating a minimum of 1 scfm to that vacuum.

a. Eductor/jet/venturi(s) shall be provided to generate 29.5 inHg for each temperature control zone. Eductor/jet/ venturi(s) shall exhaust external to the console and shall be muffled to a maximum of 45 dB aweighted sound pressure level. They shall derive their motive power from shop compressed air of a minimum of 60, to a maximum of 150 psig. Compressed air shall be accepted into the console utilizing a male air coupling half, conforming to MIL-C-4109, style 2, type II, class A, 1/4 inch steel. Filters shall be provided to filter water and oil from shop supplied air.

b. An electric (oil-less) vacuum pump shall be provided which is internal to the repair set and shall operate using the same power source.

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c. The vacuum shall be manually adjustable from 0 inHg to maximum output.

d. An independent vacuum outlet shall be supplied for each temperature control zone. Female vacuum connectors conforming to ISO 7241-1, series B, vacuum rated, shall be mounted to the console. Outlets shall be located so as not to interfere with the operation of the console or block visual displays.

e. An external male vacuum source port, conforming to ISO 7241-1, series B, vacuum rated, shall be provided so that vacuum can be applied from an alternative source.

f. A gage shall be provided to display the vacuum source pressure with accuracy of ±2% of full-scale range. This gage shall be of the flush-mounted dial type with a minimum dial diameter of 2.5 inches, shall conform to ANSI/ASME B40.1 and shall be mounted for quick removal.

g. Two vacuum gage and port assemblies (with disposable filters) shall be provided to monitor the vacuum level at the repair area.

3.4.1.6 Heater blankets. The repair system shall be supplied with heater blankets meeting the following criteria.

3.4.1.6.1 General. All heater blankets provided shall meet these requirements:

a. The heater blankets shall be of the electrical-resistance type and have a maximum operating temperature of not less than 600 °F.

b. The heater blankets shall be designed for the direct application of heat without temperature gradients greater than ±20 °F measured from the center of the blanket to all points one inch from the edge of the blanket. This temperature gradient shall be verified as specified in 4.6.3.12.

c. The blankets shall plug into the controller and shall have 10 feet nominal power leads. Blankets shall utilize a straight plug which conforms to MS 3451, class W, size 22-28, style P. Blanket connectors shall utilize pins A and F for 110 volt operation and pins C and D for 220 volt operation.

d. The blankets shall be flexible enough to perform a repair on a structure with a 1 inch radius.

e. The blankets shall not have an integral vacuum.

f. Thermocouples shall not be integral to the blanket.

g. The blankets shall be capable to withstand the environmental conditions which include oils, solvents, temperature extremes, vacuum, etc., and still perform their intended function.

h. The heater filament shall be located to within 1/8 inch to the edge of the blanket so that a 1/4 inch gap will exist between filaments when two blankets are used side-by-side (excluding lead connection side).

i. All connecting wires and leads shall be designed to eliminate air leak paths (i.e., fused and non-porous insulation and covers) during vacuum bagging operations and shall be installed using strain relief techniques.

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j. A cold tab shall be employed that allows lead connections to the blanket to be separate from the heating area. The cold tab dimensions shall not be considered as part of the blanket size.

k. Blankets shall be marked with "THIS SIDE UP" on the appropriate side and the following information: voltage, wattage, watt density, maximum rated temperature, date of manufacture and name of manufacturer.

3.4.1.6.2 Initial blanket delivery. Each repair set delivered shall be supplied with 2 each, 8 inch-, 10 inch-, and 12 inch-square and 1 each, 14 inch- and 16 inch-square heater blankets. The heater blankets shall operate at 115V input and shall be rated at an overall average of 5 watts per square inch. To achieve the temperature gradient criteria as specified in 3.4.1.6.1.b, variable watt density throughout the blanket is permitted.

3.4.1.6.3 Additional blankets. Additional blankets may be purchased as specified (see 6.2). These additional blankets shall meet the criteria specified in 3.4.1.6.1 as well as the following criteria:

a. The blankets shall operate at voltage inputs of 110V or 220V from the temperature controller and shall be rated at 2 to 5 watts per square inch dependent upon application. User must specify within these ranges.

b. Shape and size of blanket must be specified.

3.4.1.7 Cables and connectors. The repair set shall be provided with a 10 foot input power cable with the capability to meet the power requirements specified in 3.2.1.1. The power cable shall have a 30 amp 250 volt plug that plugs into the power source (an adapter "pigtail" shall be provided to facilitate 110v 15 amp outlets). The power connection to the console shall be a 90 degree plug which conforms to MS3408, class D, size 20-19, style S. The repair set shall also be provided with 2 sets of vacuum hoses, each set having two, 20 foot- and one, 10 foot-long hose, with the capability of connecting the 20 foot and 10 foot hoses together, and to the vacuum bag connections. Each hose shall be fabricated using reinforced vacuum hose with 3/8 inch inside diameter. Hoses shall have male and female vacuum connectors conforming to ISO 7241-1, series B, vacuum rated, attached at opposite ends. Ten temperature J-type thermocouples each 10 feet long and 20 thermocouple extension cables, each 20 feet long shall be provided. In addition, two sets of blanket extension cords shall be provided, each set having two 20 foot extension cords. Heater blanket extension cords shall utilize a straight plug which conforms to MS3456, class W, size 22-28, style P on one end and a cable connecting socket conforming to MS3451, class W, size 22-28, style S. Blanket connectors shall utilize pins A and F for 110 volt operation and pins C and D for 220 volt operation.

3.4.1.8 Expendables. All expendables such as paper, ribbons, filters etc., prescribed in this document shall be commercially available from at least two sources.

3.4.2. Physical characteristics

3.4.2.1 Console. All operating controls shall be located on a repair set console in a location convenient to the operator. Power-in connection will be installed on the left side of the console, blanket

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power-out shall be located on the right side of the console. Thermocouple, shop air inlets, and vacuum outlets shall be located as not to interfere with the operation of the console or block visual displays. Dials and switches shall be functionally arranged and clearly identified. All connections and plug-ins shall be on the repair set console face. The console shall include the features specified in 3.4.1. The temperature control zones shall be numbered, not color-coded.

3.4.2.2 Size and weight. The repair set shall be of the minimum practical size and weight consistent with the performance requirements of this document. The repair set shall not weigh more than 60 pounds per container and consist of no more than 2 containers. Containers shall not exceed the following dimensions: 36 inches length; 24 inches width; and 12 inches height (when in "closed" configuration).

3.4.2.3 Surfaces. All repair set surfaces shall be clean, smooth finished in accordance with MIL-STD-808, and the edges shall be rounded or beveled except where required to perform a necessary function. The exterior surfaces of the portable case shall be O. D., color number 34087 and the console shall be gray, color number 36492 conforming to FED-STD-595. Lettering shall be in accordance with FED-STD-595.

3.4.2.4 Explosion proofing. The repair set shall be designed to meet the explosion proof requirements of MIL-STD-810, method 511.3, procedure I.

3.4.2.5 Plug-in electronics. All electronic printed circuit boards and motherboards shall be conformal coated in accordance with MIL-C-28809. In addition, all electronic printed circuit boards shall conform to MIL-P-13949, MIL-P-50884 and MIL-P-55110.

3.4.2.6 Container. The containers shall be style A enclosure for temperature controllers, monitoring devices, vacuum source, alarms, and temperature indicators/recorders, and style T enclosure for cables, hoses and heater blankets. Both styles shall be in accordance with MIL-T-28800. Each enclosure shall provide protection for the contained equipment from the specified environmental conditions. The containers shall consist of a main case and a cover which allows operation in bench-top or portable applications when the cover is opened. When in operation the cover must be capable of being locked in the open position. All external electrical connections and all operating controls shall be on the console. Any ventilation openings shall be located on the console panel. The containers shall be provided with tension latches of the quick opening type which shall be protected from damage by contact with other objects. Spring wire shall not be used as a mechanical hold down device. If two container repair sets are employed, the containers shall nest for shipment and storage.

3.4.2.6.1 Pressure equalizing valve. Each container shall be provided with a pressure equalizing valve. The valve shall be accessible when the container is closed, and shall be recess mounted so that it does not protrude beyond the container.

3.4.2.6.2 Handles. The repair set shall be provided with handles conforming to the style T container requirements specified in MIL-T-28800.

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3.4.3 Reliability. The reliability for the repair set shall be as defined in MIL-STD-781. The repair set shall have a minimum acceptable mean-time-between-failure (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.4.4 Useful life. As a design goal the repair set shall be capable of meeting the performance requirements of this specification when exposed to the natural and induced environments for a minimum of 20 years calendar time or 50,000 operating hours.

3.4.5 Maintainability. Maintenance shall be accomplished in the field shops using standard tools and skills with procedures furnished by the manufacturer. A mean time to repair (MTTR) as defined in MIL-STD-721 shall be 2 hours or less. A maximum time to repair shall be 4 hours for 98% of the repair set components.

3.4.5.1. Disassembly provisions. Provisions shall be made for ready disconnection, removal and replacement of major assemblies or components, which may require replacement, major repair or overhaul. Whenever possible, this disassembly shall not require removal of other components or assemblies. Pilot guides, slides and other similar features shall be used where they will facilitate replacement of assemblies or components. The design and selection of disconnecting means for wires, tubing, control cables, etc., shall be such that incorrect assembly is avoided. Disconnection points shall be suitably identified.

3.4.6 Environmental conditions. The repair set shall be designed and constructed to operate in the exposed environmental conditions specified herein with no degradation of materials or performance when tested as specified herein.

3.4.6.1 Natural environment

3.4.6.1.1 Temperature. The repair set shall withstand temperature extremes of -65 to 180 °F as specified in MIL-STD-810, methods 501.3 and 502.3. Minimum acceptable operating conditions of the repair set shall include temperature extremes of 32 to 120 °F.

3.4.6.1.2. Humidity. The repair set shall withstand relative humidity extremes up to 95% over the temperature range as specified in MIL-STD-810, method 507.3, procedure III.

3.4.6.2 Induced environment

3.4.6.2.1 Operation in tilted position. The repair set shall be capable of functioning satisfactorily while its plane of operation is 90° or less from the horizontal plane in any direction.

3.4.6.2.2 Vibration. The repair set induced vibration shall be as described in MIL-STD-810, method 514.4, category I with 90 minutes per axis test duration.

3.4.6.2.3 Shock. The repair set shock shall be as described in MIL-STD-810, method 516.4, procedure IV (transit drop).

3.4.7 Transportability. The repair set shall be designed to remain functional after normal handling in any position during transportation by air, truck or rail without need for recalibration. The repair set shall require no special packaging for transportation.

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3.4.8 Calibration. The repair set shall require calibration no more than once every year. Mean time to calibrate shall be 2 hours or less.

3.5 Identification of product. All equipment, unit assemblies, subassemblies, modules and replaceable parts shall be marked for identification in accordance with MIL-STD-130.

3.5.1 Operating and maintenance instructions. Operating, maintenance instructions, and parts listing for the repair set shall be provided as follows:

a. An operating instruction plate shall be attached on the repair set positioned for easy reviewing from the operator's position. Instructions shall include abbreviated step-by-step operating procedures from power-on through power-off.

b. A detailed operating and maintenance instruction booklet shall be provided. To be included are detailed operating procedures, maintenance and calibration procedures, vendor data, illustrated parts breakdown (IPB), parts listing with suppliers, and instructions defining removal of components for repair. Illustrations may be included for clarity.

3.5.2 Conductor identification. All conductors shall be identified at each connection point with the code numbers shown on the wiring diagram. Terminal numbers which shall be the same as wire numbers shall be shown on all terminal board points used.

3.5.3 Safety markings. Warning and other markings shall be applied to the equipment and its individual components to aid personnel in avoiding potentially hazardous conditions in accordance with MIL-STD-1472. Additional safety markings, if required, shall be called out in the procurement document.

3.6 Workmanship. The repair set shall be free of defects which may affect strength, durability, safety or serviceability. Of particular interest are quality of welding, finish or exposed edges of nonmetallic materials, and preparation of surfaces before painting. The repair set shall be smooth and free of sharp or jagged edges.

3.6.1 Fastening devices. All screws, pins, bolts and similar parts shall be installed with adequate means to prevent loss of proper torque and adjustment. All such parts, when subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.6.2 System safety. Hazards to personnel and equipment shall be minimized by the application of good design principles in system safety. The safety design criteria of MIL-STD-454 shall be adhered to during the design and construction of the equipment.

3.6.2.1 System safety features. The repair set shall be accepted for use around a fueled aircraft on the flightline or in the hangar. The repair set and accessories shall conform to the requirements of MIL-STD-810, method 511.3, procedure I.

3.6.2.1.1 Grounding requirement. The repair set shall incorporate a grounding system that is compatible with all components of the system. The grounding system shall prevent ground loops

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and ground returns which are common to signal and power circuits, minimize electromagnetic interference (EMI), provide a single exterior grounding point and protect personnel from electrical hazards. All exposed metal parts, shields, control shafts, etc., shall be grounded to the chassis. Slip joints, bearings, and bearing surfaces shall not be used as part of the ground circuit or power return. All noncurrent-carrying metal parts shall be grounded by noncurrent-carrying conductors. The grounded circuit conductor (neutral) shall be insulated from the grounding conductors and from equipment enclosures and other grounded parts. Neutral power conductors shall be white and equipment grounding wires shall be green.

3.6.2.1.2 Ground fault protection. An overload relay/ground fault interrupter shall be provided for the repair set. This feature must be capable of operation in an explosive atmosphere as outlined in MIL-STD-810.

3.6.2.1.3 Electrical shock. The danger to personnel, from electrical shock, shall be avoided by suitable interlocks, grounding means, enclosures or protective devices conforming to the safety design criteria specified in MIL-STD-454. All contacts, terminals and like devices shall be provided with barriers or guards to prevent personnel from electrical shock.

3.6.2.1.4 Fire prevention. Fire prevention measures such as separation of hazardous substances from heat sources, incorporation of spark arresters, and suitable vents and drains shall be provided to protect personnel from fire and explosions.

3.6.2.1.5 Guards. Suitable guards shall be provided over moving parts with which the operator may come in contact or entangled. Guards or insulation shall be provided for areas or components with normal operating temperatures of 110 °F or higher (excluding blankets). All guards shall be designed so as to allow inspection of components within.

3.6.3 Human performance. The system shall be designed and manufactured to minimize the possibility of human error in accordance with MIL-STD-1472 to ensure human/machine compatibility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as

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part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document.

4.4 First article inspection. First article inspection shall be conducted on two aircraft repair sets. The order of testing and test sequence shall be established by the contractor so as to minimize cost and schedule.

4.4.1 Test measure (instrument) errors. Test measurement (instrument) errors shall not exceed the limits specified below:

Temperature ±5°	Time ±2%
Acceleration ±5%	Relative Humidity ±5%
Pressure ±2%	Weight ±1%

4.4.2 Special tests and examinations. Special tests and examinations, included in table I, shall be conducted to verify first article compliance.

4.4.3 First article inspection report. A first article inspection report shall be prepared by the contractor in accordance with MIL-STD-831 and submitted to the procuring activity.

4.5 Quality conformance inspection. Each production article shall be subjected to the examinations listed in table II to assure compliance with the requirements of section 3 prior to delivery to the procurement agency.

4.5.1. Rejection and retest. When an item selected from a production run fails to meet the specification requirements, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall provide a letter to the procuring agency explaining the cause of failure and the action to be taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring agency of all the defects likely to be found and the method of correcting them. Correction of defects shall be at contractor expense.

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TABLE I. First article inspections.

Requirement	Paragraph	Examination Type	Paragraph
Steady-state conditions	3.2.1.1.1	Demonstration	4.6.2.5 thru 4.6.2.7
Transient-state conditions	3.2.1.1.2	Test	4.6.3.7 thru 4.6.3.9
Interruption of power	3.2.1.1.3	Test	4.6.3.10
Heater blankets	3.4.1.6	Test	4.6.3.12
Cables and connectors	3.4.1.7	Demonstration	All
Console	3.4.2.1	Demonstration	All
Explosion proofing	3.4.2.4	Test	4.6.3.11
Plug-in electronics	3.4.2.5	Inspection	4.6.1.2
Container	3.4.2.6	Inspection	4.6.1.2
Pressure equalizing valve	3.4.2.6.1	Inspection	4.6.1.2
Handles	3.4.2.6.2	Inspection	4.6.1.2
Reliability	3.4.3	Analysis	4.6.1.1
Useful life	3.4.4	Test	4.6.3.2
Maintainability	3.4.5	Analysis	4.6.1.1
		Demonstration	4.6.2.5 thru 4.6.2.8
Disassembly Provisions	3.4.5.1	Demonstration	4.6.2.5 thru 4.6.2.8
		Inspection	4.6.1.2
Temperature	3.4.6.1.1	Test	4.6.3.5
			4.6.3.6
Humidity	3.4.6.1.2	Test	4.6.3.4
Operation in tilted position	3.4.6.2.1	Demonstration	4.6.2.4
Vibration	3.4.6.2.2	Test	4.6.3.3
Shock	3.4.6.2.3	Test	4.6.3.13
Transportability	3.4.7	Inspection	4.6.1.2
		Analysis	4.6.1.1
Calibration	3.4.8	Demonstration	4.6.2.6
O&M instructions	3.5.1	Demonstration	All

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TABLE I. First article inspections. - Contd

Requirement	Paragraph	Examination Type	Paragraph
System safety	3.6.2	Analysis	4.6.1.1
System safety features	3.6.2.1	Test	4.6.3.11
Ground fault protection	3.6.2.1.2	Inspection	4.6.1.2
Fire prevention	3.6.2.1.4	Inspection	4.6.1.2
		Analysis	4.6.1.1
Human performance	3.6.3	Demonstration	All
		Analysis	4.6.1.1

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TABLE II. Quality conformance inspections.

Requirement	Paragraph	Examination Type	Paragraph
Power	3.2.1.1	Inspection	4.6.1.2
		Demonstration	4.6.2.1/.2
			4.6.2.3/.4
Operating conditions	3.2.1.2	Demonstration	4.6.2.7
Materials, parts and processes	3.3.1	Inspection	4.6.1.2
Welding, brazing, and soldering	3.3.1.1	Inspection	4.6.1.2
Standard parts	3.3.1.2	Inspection	4.6.1.2
Magnesium	3.3.1.3	Inspection	4.6.1.2
Dissimilar metals	3.3.1.4	Inspection	4.6.1.2
Fungus resistance	3.3.1.5	Analysis	4.6.1.1
Corrosion resistance	3.3.1.6	Analysis	4.6.1.1
Measuring and Indicating devices	3.3.1.7	Inspection	4.6.1.2
Temperature controller	3.4.1	Demonstration	4.6.2.1 thru
			4.6.2.4
			4.6.2.5/.7
			4.6.2.8/.9
		Inspection	4.6.1.2
		Analysis	4.6.1.1
Controller programability	3.4.1.1	Demonstration	4.6.2.2/.3
			4.6.2.4/.7
			4.6.2.8/.9
Thermocouples	3.4.1.2	Inspection	4.6.1.2
Auxiliary temperature indicator	3.4.1.3	Demonstration	4.6.2.3
		Inspection	4.6.1.2
		Analysis	4.6.1.1
Temp/vacuum/deviation alarm recorders	3.4.1.4	Demonstration	4.6.2.1/.2
			4.6.2.3/.4
			4.6.2.5/.7
			4.6.2.8/.9
		Inspection	4.6.1.2
		Analysis	4.6.1.1

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TABLE II. Quality conformance inspections.- Contd

Requirement	Paragraph	Examination Type	Paragraph
Vacuum	3.4.1.5	Inspection	4.6.1.2
		Demonstration	4.6.2.1/.2
			4.6.2.3/.4
			4.6.2.5/.7
			4.6.2.8/.9
		Analysis	4.6.1.1
Heater blankets	3.4.1.6	Analysis	4.6.1.1
		Inspection	4.6.1.2
Cables and connectors	3.4.1.7	Inspection	4.6.1.2
Expendables	3.4.1.8	Analysis	4.6.1.1
Console	3.4.2.1	Inspection	4.6.1.2
Size and Weight	3.4.2.2	Test	4.6.3.1
Surfaces	3.4.2.3	Inspection	4.6.1.2
Container	3.4.2.6	Inspection	4.6.1.2
Pressure equalizing valve	3.4.2.6.1	Inspection	4.6.1.2
Handles	3.4.2.6.2	Inspection	4.6.1.2
Identification of product	3.5	Inspection	4.6.1.2
O&M Instructions	3.5.1	Inspection	4.6.1.2
Conductor identification	3.5.2	Inspection	4.6.1.2
Safety markings	3.5.3	Inspection	4.6.1.2
Workmanship	3.6	Inspection	4.6.1.2
Fastening devices	3.6.1	Inspection	4.6.1.2
Grounding requirement	3.6.2.1.1	Inspection	4.6.1.2
Electrical shock	3.6.2.1.3	Inspection	4.6.1.2
Guards	3.6.2.1.5	Inspection	4.6.1.2

4.6 Methods of Inspection.

4.6.1 Analysis and Inspection.

4.6.1.1 Analysis. The requirements of section 3 shall be verified by use of contractor design data, contractor and subcontractor specifications, drawings, handbook, etc.

4.6.1.2 Inspection. The requirements of section 3 shall be verified by inspection. Inspection is described as reviews and comparison of drawings, documents, subcontractor inspection records, actual hardware, etc., which do not require mathematical calculations to arrive at a conclusion.

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4.6.2 Demonstrations. The requirements of section 3 shall be verified by repair system demonstrations of actual repair scenarios to DOD procurement or technical representatives.

4.6.2.1 Power demonstration. The repair set shall be plugged into a 110V AC/60 Hz power supply and turned on. Each temperature controller shall be programmed, in accordance with paragraph 3.4.1.1, to operate separately from room temperature to 250 °F at 8 °F/min, hold at 250 °F for 10 minutes and cool down 8 °F/min to room temperature. A 12 inch square heater blanket shall be used and a vacuum gage and one thermocouple (control) shall be attached and placed on a 24 inch square 2024 aluminum plate, 0.125 inch thick, conforming to QQ-A-250, during the process. Insulate as necessary to achieve and maintain the 250 °F temperature. The plate shall be vacuum bagged. During the program run for each controller, the vacuum system shall provide at least 25 inches of mercury pressure to the plate and the temperature/vacuum/alarm deviation recorders shall be in operation during the entire process. The controller shall display the programmed cure cycle and the real-time actual cure cycle in accordance with 3.4.1.n. The process for each controller shall be repeated using a 220V AC/50 Hz power input. A completed cycle with the proper visual displays, a hardcopy printout, no vacuum/temperature/deviation alarms sounding, and an end of cycle alarm sounding shall constitute acceptance.

4.6.2.2 Temperature control. The repair set shall be plugged into a 110V AC/60 Hz power supply and turned on. Each temperature controller shall be programmed, in accordance with 3.4.1.1, to operate separately from room temperature to 200 °F, held at 200 °F for 2 minutes, rise from 200 °F to 350 °F, held at 350 °F for 4 minutes, rise from 350 °F to 500 °F, held at 500 °F for 5 minutes, and then cooled down to 120 °F (indicating cooldown complete). All ramp and cool rates shall be 5 °F/min. A 12 inch square heater blanket shall be used and vacuum gage and one thermocouple shall be attached and placed on a 24 inch square 2024 aluminum plate, 0.125 inch thick, conforming to QQ-A-250, during the process. A calibrated and independent temperature monitoring device shall be used to evaluate the controller thermocouple readings under the heating blanket. Insulate as necessary to achieve and maintain the repair temperatures. The plate shall be vacuum bagged. During the program run for each controller, the vacuum shall provide at least 25 inches of mercury pressure to the plate and the temperature/vacuum/ deviation alarm recorders shall be in operation during the entire process. The controller shall display the programmed cure cycle and the real-time actual cure cycle in accordance with 3.4.1.n. The process shall be repeated for each controller using a 220V AC/50 Hz power input. A completed cycle with the proper visual displays, a hardcopy printout, no vacuum/ temperature/deviation alarms sounding, and an end of cycle alarm sounding shall constitute acceptance.

4.6.2.3 Auxiliary temperature indicator. The repair set shall be subjected to the demonstration described in 4.6.2.2, 110V AC/60 Hz only for both controllers. Four auxiliary thermocouples shall be placed under the blanket in the vicinity of the control thermocouple. A calibrated and independent temperature monitoring device shall be used to evaluate the auxiliary thermocouple readings as well as the control thermocouple reading. During the program run, the auxiliary thermocouple selector switch shall be used to display each of the thermocouples and compared to the control temper-

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ature. Also, periodic printouts of the thermocouple readings shall be made. Auxiliary thermocouple temperature readings within ±5 °F shall constitute acceptance.

4.6.2.4 Operation in tilted position. The repair set shall be subjected to the demonstration described in 4.6.2.2, 110V AC/60 Hz only with the exception that the repair set shall be tilted 90 degrees in each direction about the longitudinal and lateral direction.

4.6.2.5 Accessibility demonstration. The repair set shall have all removeable components (e.g. vacuum filters, temperature controllers, gages, recorder) removed with minimal tooling and replaced. The repair set shall then be turned on and subjected to the demonstration as described in 4.6.2.2, 110V AC/60 Hz condition only. Successful completion of the cycle, speed of removal process, and ease of operation shall constitute acceptance.

4.6.2.6 Calibration demonstration. The contractor shall follow the developed calibration procedures, required under 3.4.8, and calibrate the repair set using only the contractor supplied instructions within 2 hours.

4.6.2.7 Operating conditions demonstration. The repair set shall be subjected to the demonstrations in 4.6.2.1 and 4.6.2.2, 110V AC/60 Hz condition only, by personnel wearing chemical/biological warfare gloves conforming to MIL-G-43976.

4.6.2.8 Maintainability demonstration. The repair set shall have all removable components (e.g. temperature controllers, gages, temperature/vacuum recorders) removed and interchanged to verify maintainability times in 3.4.5. The repair set shall then be subjected to the demonstration described in 4.6.2.2, 110 AC/60 Hz condition only.

4.6.2.9 Temperature/vacuum deviation alarm demonstration. The repair set shall be subjected to the demonstration in 4.6.2.2, 110 AV/60 Hz condition only, with the following exceptions: (1) the low limit vacuum alarm shall be set a 20 inHg (set high limit at 29 inHg). During the run, temporarily open the seal to the vacuum bag such that vacuum is lost and the alarm is trigerred. Reseal to verify alarm resets itself and verify that the printer records the deviation; (2) the high limit vacuum alarm shall be set a 17 inHg (set low limit at 0 inHg). Initiate the run without a good seal on the bag. After approximately 10 minutes, seal the vacuum bag properly. The alarm should sound and printer should record the deviation. Open the vacuum bag seal such that alarm resets itself and printer records the deviation correction properly; (3) external devices shall be used to cause excessive temperature deviations to occur. During the program run, at both ramp and dwell stages, temperature deviations shall occur to cause the alarms and printer to activate. Measures shall then be taken to correct the deviation causes and resume normal operation of the program. Sounding of the audible alarm, illumination of the visible alarm, resetting of the alarms when the deviation is corrected, and printout of the alarm messages and corrective actions in accordance with 3.4.1.4 shall constitute acceptance for all three cases.

4.6.3 Tests. The requirements of section 3 shall be verified by tests of an actual repair set. Failure criteria shall be as specified in MIL-STD-810.

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4.6.3.1 Weight. The repair set shall be weighed. Each container (in operational configuration) shall be 60 pounds or less to constitute acceptance.

4.6.3.2 Useful life. To validate the repair set for repeated use, the repair set shall be subjected to the demonstrations described in 4.6.2.2 repeatedly for 4 complete cycles for each power condition.

4.6.3.3 Vibration. The repair set shall be subjected to the vibration tests as specified in category 1, method 514.4 of MIL-STD-810 to the levels as shown on figure 1 for 90 minutes per axis. After the vibration test, the repair set shall be operated as described in 4.6.2.2, 110 AC/60 Hz condition only.

4.6.3.4 Humidity. The repair set shall be subjected to the humidity test as specified in procedure III, method 507.3 of MIL-STD-810. The repair set shall be operated before exposure and near the end of every cycle as described in 4.6.2.2, 110 AC/60 Hz condition only.

4.6.3.5 High temperature. The repair set shall be subjected to the high temperature tests as specified in procedures I and II, method 501.3 of MIL-STD-810. The high storage temperature shall be 180 °F, and the high operating temperature shall be 120 °F. The repair set shall be operated at 120 °F as described in 4.6.2.2, 110 AC/60 Hz condition only.

4.6.3.6 Low temperature. The repair set shall be subjected to the low temperature test as specified in procedures I and II, method 502.3 of MIL-STD-810. The low storage temperature shall be -60 °F, and the low operating temperature shall be 32 °F. The storage temperature shall be maintained for 24 hours. The repair set shall be operated at 32 °F as described in 4.6.2.2, 110 AC/60 Hz condition only.

4.6.3.7 Voltage and frequency variation. The repair set response to the input voltage and frequency variation shall be tested as follows:

Step 1. Connect the power cable of the repair set to a variable voltage and frequency source.

Step 2. Set the output of the variable source at 115V, 60 Hz.

Step 3. Performance of the repair set outside the limits required in 4.6.2.2, 110 AC/60 condition only shall constitute a failure of this test.

Step 4. Repeat steps 2 and 3 for voltages and frequencies from the variable source as shown in table III, using the upper and lower tolerance limit values of columns (a) and (b). The voltage and frequency shall be maintained for at least 15 minutes at each level specified.

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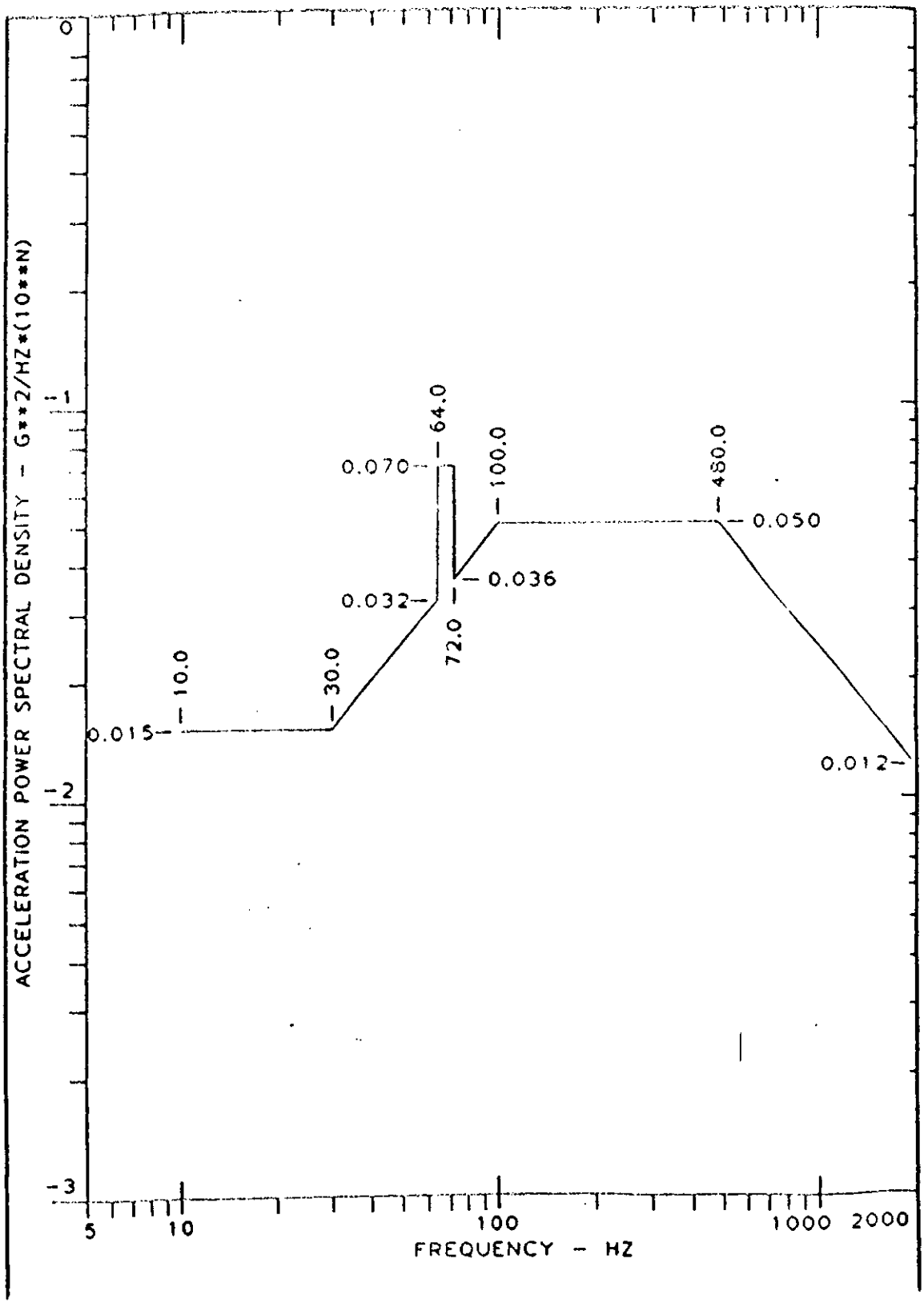


FIGURE 1. Air transport vibration (90 minutes per axis, all three axes).

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TABLE III. Variable frequency power supply outputs.

Frequency	(a) Minimum Voltage	(b) Maximum Voltage
47	103.5	126.5
53	103.5	126.5
57	103.5	126.5
63	103.5	126.5

4.6.3.8 Voltage transients. The repair set performance during voltage transients shall be tested as follows:

Step 1. Connect the power cable of the repair set to a 60 Hz variable voltage source.

Step 2. Set the output of the source to the maximum voltage for the repair set as indicated in table III.

Step 3. Quickly change the source voltage to a transient value of 149.5 volts, hold for approximately 0.5 seconds, and quickly return to the starting voltage of step 2. Monitor repair set performance during and after the transient for compliance with 4.6.2.2, 110 AC/60 Hz condition only. Performance which is outside the specified limits during the transient for longer than 30 seconds, shall constitute a failure of this test.

Step 4. Quickly change the source voltage to a transient value of 161 volts, hold for approximately 0.5 seconds, and quickly return to the starting voltage of 126.5 volts. Monitor repair set performance during and after the transient for compliance with 4.6.2.2, 110 AC/60 Hz condition only. Performance which is outside the specified limits after the transient has passed shall constitute a failure of this test, except that any externally accessible fuses that have failed may be replaced if the specified performance is restored by replacement of the fuse(s).

Step 5. Repeat steps 3 and 4 four more times at 2-minute intervals.

Step 6. Set the output of the source to a starting voltage of 103.5 volts.

Step 7. Repeat step 3 with a transient value of 80.5 volts and the starting voltage specified in Step 6.

Step 8. Repeat step 4 with a transient value of 69 volts and the starting voltage specified in Step 6.

Step 9. Repeat steps 7 and 8 four more times at 2-minute intervals.

4.6.3.9 Frequency transients. The repair set performance during frequency transients shall be tested as follows:

Step 1. Connect the power cable of the repair set to a 115 Volt variable frequency source. The output voltage shall be the operating voltage specified for the repair set.

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Step 2. Set the output of the source to a starting frequency of 47.5 Hz.

Step 3. Quickly change the source frequency to a transient value of 45 Hz, hold for approximately 0.5 seconds, and quickly return to the starting frequency of 47.5 Hz. Monitor repair set performance during and after the transient for compliance with 4.6.2.2 ,110 AC/60 Hz condition only. Performance which is outside the specified limits during the transient for longer than 30 seconds, shall constitute a failure of this test.

Step 4. Quickly change the source frequency to a transient value of 42.5 Hz, hold for approximately 0.5 second, and quickly return to the starting frequency of 47.5 Hz. Monitor the repair set performance during and after the transient for compliance with 4.6.2.2, 110 AC/60 Hz condition only.

Step 5. Repeat steps 3 and 4 four more times at 1-minute intervals.

Step 6. Set the output frequency of the source to a starting frequency of 57 Hz.

Step 7. Repeat step 3 with a transient frequency of 54 Hz and a starting frequency of 57 Hz.

Step 8. Repeat step 4 with a transient frequency of 54 Hz and a starting frequency of 57 Hz.

Step 9. Repeat steps 7 and 8 four more times at 1-minute intervals.

Step 10. Set the output frequency of the source to a starting frequency of 63 Hz.

Step 11. Repeat step 3 with a transient frequency of 66 Hz and a starting frequency of 63 Hz.

Step 12. Repeat step 4 with a transient frequency of 69 Hz and a starting frequency of 63 Hz.

Step 13. Repeat steps 11 and 12 four more times at 1-minute intervals.

4.6.3.10 Interruption of power. The repair set shall operate as described in 4.6.2.2, 110 AC/60 Hz condition only and shall be turned off for 2 minutes. The repair set shall be turned on and automatically resume the cycle.

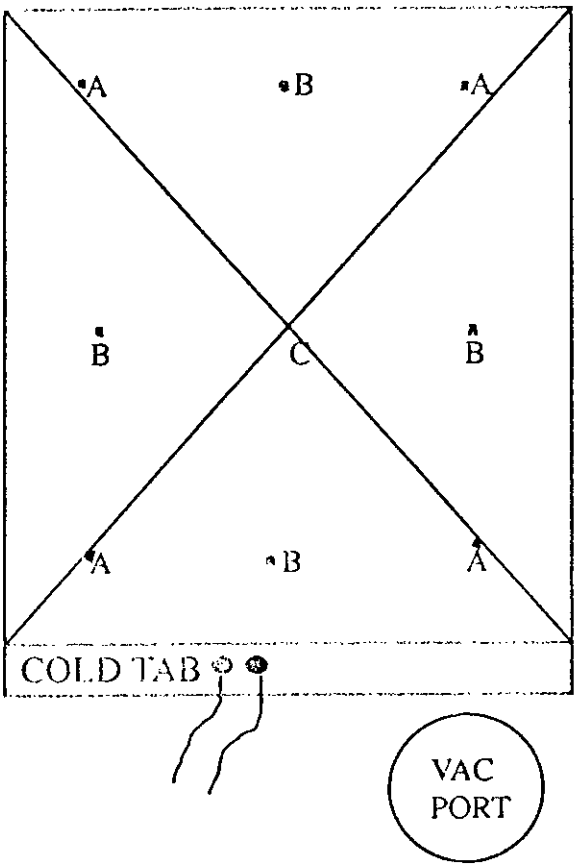
4.6.3.11 Explosion proofing. The repair set shall be subjected to the tests as specified in procedure I, 511.3 of MIL-STD-810. The operation shall be in accordance with 4.6.2.2, 110 AC/60 Hz condition only.

4.6.3.12 Heater blanket. Each heater blanket delivered shall be subjected to a temperature gradient test for compliance with 3.4.1.6.1.b. The test shall be conducted as follows:

a. A composite panel shall be used to determine temperature variations. The panel shall consist of 16 plies of unidirectional graphite/epoxy material fabricated with a [0, 45, 90, -45]2S ply layup pattern. The panel shall have dimensions such that 2.0 inches overlaps the blanket being tested on each side.

b. J-type thermocouples shall be taped down on the panel surface according to the layout on figure 2. (Outlining the blanket onto the panel is helpful in performing this step.)

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A – Thermocouple located at diagonal of heating area of blanket 1.0 inch from each edge.
B – Thermocouple located 1.0 inch from edge and centered edge to edge of heating area.
C – Control thermocouple located at center of heating area of blanket.

FIGURE 2. Heating blanket test configuration.

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c. The panel, with thermocouples attached, shall be placed on top of 15 plies of insulation (Richmond RC-3000-10 Breather or equivalent).

d. Place the blanket on the panel, ensuring the blanket is positioned properly with respect to the thermocouples. Place an additional 15 plies of insulation (Richmond RC-3000-10 Breather or equivalent) on top of the blanket.

e. Attach the thermocouple leads to a thermocouple monitoring device. Ensure the center thermocouple, labeled "C", is used as the control thermocouple.

f. Vacuum bag the blanket/panel assembly. Place the vacuum port along the "COLD TAB" side of the blanket.

g. Seal the vacuum bag and apply vacuum. Ensure a vacuum level of 18 inHg minimum is obtained.

h. Ramp the temperature to 350 °F with the temperature controller. When the temperature reaches 350 °F, allow the temperatures to stabilize for 1 minute, then record all thermocouple readings. Hold the temperature at 350 °F for 60 minutes, during which time recordings of all thermocouples shall be taken at 30 minutes and 60 minutes.

i. Criteria for blanket acceptance shall be a maximum temperature difference of ±20 °F for thermocouple "C" to thermocouple "A" and "B".

4.6.3.13 Shock. The repair set shall be subjected to the shock tests as specified in procedure IV (transit drop), method 516.4 of MIL-STD-810 to the levels specified in table 516.6-II. After the shock test, the repair set shall be operated as described in 4.6.2.2, 110AC/60 Hz condition only.

5. PACKAGING

5.1 General. Cases covered by this specification are primarily designed to protect the equipment during service usage, shipment, and storage. Multiple container repair sets shall be banded together for shipment.

5.2 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended Use. The repair set covered by this specification is intended to be used for the repair of aircraft composite and metal bonded structures which necessitate the application of heat and pressure. The set is intended for use on the flightline or within a maintenance hangar.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specificaiton.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. When a first article is required (see 3.1 and 4.4).

d. Shape and size blanket required, initial and additional (see 3.4.1.6).

e. Additional safety markings required (see 3.4.2.3).

6.3 First article. When a first article inspection is required, the items should be selected from the first two production items. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paragraph	Data requirement title	DID number	Transmittal form
4.4.3	Service Engineering Reports	DI-S-3601A	LT
4.5	Quality Assurance Program Status Report	DI-QCIC-80112	LT

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.5 Subject term (key word) listing.

composite repair set
composite curing controller
controller
hot bonded repair set
programmable hot bonded controller

Custodian:
Air Force - 11

Review activities:
Air Force - 80, 99

Preparing activity:
Air Force - 11

Project No. 4920-F460

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, not to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractural requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-87259 (USAF)	2. DOCUMENT DATE *(YYMMDD)* 91 08 02

3. DOCUMENT TITLE
REPAIR SET, AIRCRAFT A/E24M-41

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

A. NAME	B. TELEPHONE *(Include Area Code)*	
ASD/ENES	(1) Commercial (513) 255-6295	(2) AUTOVON *(If applicable)* 785-6295
C. ADDRESS *(Include Zip Code)* WRIGHT-PATTERSON AFB OH 45433-6503	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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